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EXSEL
EXperiential and SErvice Learning

Wind Energy Monitoring System: A Cloud-Based Approach

A COURSE LEVEL PROJECT REPORT

Submitted by

III-year students of Bachelor of Technology

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in partial fulfillment of the course of

215EXS3201 / EXSEL – DESIGN-BUILD-OPERATE

Academic Year 2025 – 2026 (Even Semester)

DECLARATION

We affirm that the project work titled **"Wind Energy Monitoring System: A Cloud Based Approach"** being submitted in partial fulfilment for the award of the degree of **Bachelor of Technology** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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BONAFIDE CERTIFICATE

Certified that this project report **"Wind Energy Monitoring System: A Cloud Based Approach"** is the bonafide work of **"P. Hemalatha (99230040045), S. Praneetha (9824020001), G. Pravallika (99230040302), R. Vamsi Krishna (99230040415) M. Bhargav Rakesh (99230040667)"** who carried out the project work under my supervision.

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Submitted for the Project Viva-voce / Review held at Kalasalingam Academy of Research and Education, Krishnankoil on ……………………………

Internal Examiner **External Examiner**

ACKNOWLEDGEMENT

We are thankful for the opportunity to carry out this project under the EXSEL (EXperiential and SErvice Learning) curriculum, which enhanced our engineering skills through hands-on application and encouraged us to develop impactful solutions aligned with the Sustainable Development Goals (SDGs).

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Lastly, we appreciate the support of our faculty members, non-teaching staff, friends, and family who motivated us throughout this experiential learning journey.

PROJECT SUMMARY

Project Title	Wind Energy Monitoring System: A Cloud-Based Approach	
Project Team Members (Name with Register No)	P. Hemalatha (99230040045), S. Praneetha (9824020001), G. Pravallika (99230040302), R. Vamsi Krishna (99230040415), M. Bhargav Rakesh (99230040667).	
Guide Name/Designation	DR. A. RAMKUMAR	
Program Concentration Area	Cloud Computing and IOT	
Technical Requirements	Real-time data monitoring and Transmission	
Engineering standards and realistic constraints in these areas		
Area	**Codes & Standards / Realistic Constraints**	**Tick ✓**
Economic	The system should reduce overall cost by using cloud technology instead of expensive local infrastructure.	✓
Environmental	The system helps reduce pollution by promoting clean renewable wind energy	✓
Social	The system improves energy access and reliability	✓
Ethical	The system ensures secure and responsible use of data while protecting user privacy	✓
Health and Safety	The system helps prevent accidents by monitoring turbine conditions and detecting faults early	✓
Manufacturability	The system should be designed so that its components can be easily produced, assembled and maintained	✓
Sustainability	The system supports long-term use of clean wind energy with minimal environmental impact and efficient resource use.	✓

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ABSTRACT

Wind energy is one of the most promising renewable energy sources for sustainable power generation. However, efficient monitoring and management of wind energy systems are essential to ensure optimal performance and reliability. This project, "Wind Energy Monitoring System: A Cloud-Based Approach," focuses on developing a smart and scalable solution for real-time monitoring of wind energy systems using cloud technology.

The proposed system integrates sensors to collect key parameters such as wind speed, turbine rotation, voltage, and power output. These data are transmitted to a cloud platform through 10Tbased communication modules, enabling remote access and continuous monitoring from anywhere. The cloud infrastructure allows for data storage, visualization, and analysis, helping users identify performance trends and detect faults at an early stage.

By leveraging cloud computing, the system reduces the need for manual monitoring and improves operational efficiency. It also supports predictive maintenance, minimizing downtime and maintenance costs. The proposed solution is cost-effective, user-friendly, and highly scalable, making it suitable for both small-scale and large-scale wind energy applications.

Overall, this project demonstrates how the integration of renewable energy systems with cloud technology can enhance energy management, improve reliability, and contribute to a more sustainable future.

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LIST OF ABBERIVATION

S.No	ABBREVATION	FULLFORM
1	WECS	Wind Energy Conversion System
2	VAWT	Vertical Axis Wind Turbine
3	HAWT	Horizontal Azis Wind Turbine
4	AC	Alternating Current
5	DC	Direct Current
6	RPM	Revolution Per Minute
7	KW	Kilowatt
8	KWh	Kilowatt-hour
9	V	Voltage
10	A	Ampere
11	MPPT	Maximum Power Point Tracking
12	PWM	Pulse Width Modulation

PROBLEM STATEMENT

Increasing demand for clean and sustainable energy has led to the widespread adoption of wind energy systems. Despite their advantages, many existing wind energy installations suffer from inefficient monitoring and management practices. Traditional systems often rely on periodic manual inspections and lack continuous real-time tracking of critical parameters such as wind speed, turbine rotation speed, temperature, vibration, and power output. This results in delayed fault detection, reduced efficiency, and increased operations and maintenance costs

Moreover, the absence of intelligent data analysis and predictive mechanisms makes it difficult to optimize turbine performance under varying environmental conditions. Factors such as sudden wind fluctuations, mechanical wear and tear, and environmental stress can significantly impact energy monitored properly. Without an integrated monitoring system, it becomes challenging to ensure reliability, maximize energy output, and maximize downtime.

In addition, many existing systems are not equipped with remote monitoring capabilities, making it operators to supervise wind farms located in remote or offshore areas. This highlights the a smart, automated Wind Energy Monitoring System that leverages sensors, data analytics, and real communication technologies to continuously monitor system performance. Such a system can provide fault detection, improve efficiency, enable remote access, and support better decision-making, ultimately enhancing the overall effectiveness and sustainability of wind energy generation.

CHAPTER –I

INTRODUCTION

Global energy consumption is projected to increase by 50% by 2050, driven by rapid industrialization and population growth. Fossil fuels, which currently account for over 80% of the world's energy supply, are facing severe depletion while contributing to over 70% of global C02 emissions. In this context, wind energy has emerged as one of the most efficient, clean, and sustainable alternatives for power generation. With global installed wind capacity reaching over 1,017 GW in 2023, wind energy now accounts for approximately 7% of the world's total electricity production. Its widespread availability, zero greenhouse gas emissions during operation, and significant cost reductions of nearly 68% since 2010 make it a cornerstone of the global energy transition.

Despite its advantages, the effective utilization of wind energy depends heavily on proper monitoring and management of wind turbines. Wind turbines are often installed in remote, offshore, or geographically challengin locations where continuous manual supervision is oifficult and costly. Studies show that unplanned downtime in wind turbines can cost operators etween $10,000 to $30,000 per turbine per day.

Traditiona monitoring techniques, which rely on periodic inspection and local data collection, are inefficient and prone to delays, leading to reduced performance, undetected faults, and significantly increased maintenance costs.

Recent advancements in technology, particularly in the Internet of Things (IOT) and cloud computing, have opened powerful new possibilities for addressing these challenges. The global 10T market in the energy sector is expected to reach $35.2 billion by 2026, with an estimated 1.8 billion 10T devices deployed in the energy sector by 2025. loT-enabled sensors can sample operational data at intervals as low as I millisecond, enabling near- instantaneous fault detection and response. Cloud platforms such as AWS IOT, Google Cloud IOT, a and Microsoft Azure provide scalable infrastructure capable of handling millions of data points per second,offering centralized storage, processing, and analysis accessible from anywhere in the world.

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The proposed project, "Wind Energy Monitoring System: A Cloud-Based Approach," aims to develop an intelligent system that leverages 10T and cloud technologies for efficient monitoring and management of wind energy systems. Various sensors are deployed to measure key parameters including wind speed, turbine rotation speed, temperature, voltage, and electrical output. The collected data is transmitted through communication modules such as Wi-Fi or GSM to a cloud platform, where it is stored, processed, and visualized through interactive dashboards. The system incorporates automated alert mechanisms that notify operators via Using email when parameters exceed predefined thresholds, enabling proactive maintenance, minimizing downtime, and ensuring the continuous and optimal performance of wind energy installations.

CHAPTER-II

LITERATURE REVIEW

2.1 GENERAL

Design of Cloud Computing-Based Wind Energy Monitoring SystemWhich use the internet-based technologies for monitoring as Well managing wind turbines in a comprehensive manner. The sensors located on wind turbines reads and gathers crucial information, including the wind speed, temperature and battery voltage performance of individual turbines in real time. This data then is pushed to the cloud platforms for storage, analysis & monitoring remotely. For example, it aids in fast fault detection and energy efficiency while curtailing maintenance costs. This had the right support parameter for efficient monitoring, performance analytics, and information access from web or mobile apps around the globe.

2.2 REVIEW OF EXSITING WORKS

Qian Peng et al.[37] proposed a novel wind turbine condition monitoring method based on cloud computing in 2013. proposed system presents a data-driven approach for real-time monitoring and fault detection in wind turbines based on cloud computing technology. With the use of cloud platform bundles, operational Information from turbines is gathered and analyzed to enhance monitoring efficiency, cut down maintenance cost, and improve system stability. The remote access provides access to information on the performance of the turbine also in a continuous manner. However, the machine has limitations regarding the fusion of data originating from multiple sources for predictive maintenance in large-scale wind farms.

Selvamohan Amirtharaj and Nachiappan Rathina Prabha proposed a cloud service and SCADA based web application for monitoring renewable energy systems (2025). It employs SCADA technology for wind turbine data acquisition, Telegrafto collect various metrics, AWS S3 for scalable cloud storage, RESTAPI communication channels, InfluxDB for real-time analytics and dashboards to visualize power curves and how well the system is performing. The suggested model also allows cloud computing technologies for remote monitoring and management of data. It helps to monitor renewable power generation by analyzing the processes with real-time data and accessing operational data at any given time. However, the system is not yet ready for powerful scada limits.

In 2024, Jeyamurugan M and his team introduced a Cloud Based Wind Energy Monitoring System to improve the monitoring process of wind turbines. The system uses 10T technology and microcontrollers to collect important data such as wind speed and power generation. This information is transmitted through LAN communication and stored in the cloud, allowing users to monitor turbine performance remotely in real time.

The proposed system also helps in identifying faults quickly and improves the overall efficiency of wind energy management. However, some manual monitoring is still required, older systems consume more power, and the model needs stronger security features for large-scale industrial use.

In 2014, Dahai Zhang proposed an 10T and Cloud Based Remote Monitoring System for wind turbines to improve real-time monitoring and data management. The system uses 10T sensors to collect important parameters such as wind speed, temperature, and battery status. Technologies like GPRS and Wi-Max are used for multi-hop IP communication, allowing the collected data to be transferred directly to a cloud-based.

Dashboard This approach replaces traditional memory card and PC-based logging methods, making remote monitoring easier and more efficient. However, the system still faces challenges such as high operational costs due to human intervention, limited remote accessibility in older monitoring systems, and the absence of economic feasibility analysis for large-scale implementation.

In 2013, Yulin Si proposed a predictive cloud-based solution aimed at reducing maintenance costs in wind Farms The system uses cloud computing software to combine SCADA and sensor data for real-time anomaly detection, condition monitoring, and fault prognosis. By analyzing the collected data, the model helps identify possible failures early and supports preventive maintenance measures to improve turbine performance and reliability. This approach reduces downtime and maintenance expenses while increasing the efficiency of wind farm operations. However, older wind farms still experience high failure rates ranging from 3% to 30%, and the system requires better integration of data from multiple sources to achieve improved productivity and accurate predictions.

In 2021, A. Kusiak proposed an efficient monitoring and control system for wind energy conversion using 10T technology. The system integrates 10T devices with wind turbine generation units, SCADA systems, and cloud computing platforms to enable real-time monitoring and control of wind energy operations. By continuously collecting and analyzing turbine data, the model helps improve energy efficiency, operational reliability, and remote accessibility.

The use of cloud technology also supports faster data processing and easier management of wind farm performance. However, the study identified gaps in the effective utilization of SCADA data for advanced applications and decision-making processes in wind energy systems.

In 2022, Ramesh Kumar and his research team proposed a smart wind energy monitoring system using 10T and cloud computing technologies. The system collects real-time data such as wind speed, turbine temperature, voltage, and power output through connected sensors installed on wind turbines. The collected information is sent to a cloud platform where it can be monitored and analyzed remotely using web dashboards. This approach helps operators identify faults quickly, reduce maintenance costs, and improve the overall efficiency of wind farms. However, the system still faces challenges related to cybersecurity, handling large-scale data, and improving prediction accuracy during changing weather conditions.

CHAPTER- III

PROBLEM DEFINITION AND PROJECT OBJECTIVES

3.1 Problem Definition

Wind energy systems are increasingly being adopted as a sustainable alternative to conventional energy sources, global installed wind capacity surpassing 1,017 GW in 2023 and expected to reach 2,000 GW by 2030 according to Global Wind Energy Council. However, the efficient operation and maintenance of these systems remain challenge. Wind turbines are often installed in remote, offshore, or geographically inaccessible locations, making continuous monitoring extremely difficult using traditional methods. Studies indicate that unplanned turbine downtime can cost operators between $10,000 to $30,000 per turbine per day, highlighting the critical need for efficient and continuous monitoring solutions.

A major issue in current wind energy setups is the lack of real-time data accessibility. Important parameters such as wind speed, turbine rotation speed, voltage, current, temperature, and power output are not continuously monitored effectively analyzed in most existing installations. According to a report by the International Renewable Energy Agency published in 2022, nearly 45% of wind energy losses are attributed to inadequate monitoring and detection. Without a robust monitoring framework, energy generation efficiency drops significantly, unexpected system failures become more frequent, and maintenance costs increase substantially. The absence of centralized data storage and intelligent analysis tools further makes it difficult to track system performance over time and make informed operational decisions.

Another critical challenge is the limited scalability and prohibitively high cost of the and of is to a energy monitoring systems as SCADA. With installation costs ranging from $50,000 to $500,000 per unit, SCADA systems are entirely for small-scale or rural wind energy installations, particularly in developing countries where over 760 million people still lack access to reliable electricity according to the World Bank. A 2022 survey by the Renewable Energy Policy Network found that more than 55% of small-scale wind energy operators in developing regions reported that the of monitoring infrastructure was the primary barrier to efficient system management. Furthermore, without pro integration of modem technologies such as IOT, cloud computing, and machine learning, existing systems fail to leverage the benefits of automation, remote monitoring, and predictive maintenance.

The financial impact of these monitoring deficiencies is considerable. Research published by the European Wind Energy Association estimates that inadequate monitoring and maintenance practices result in an average annual loss to 10% across wind farm installations globally. In addition, reactive maintenance approaches, which are still widely practiced due to the absence of smart monitoring systems, cost on average three times more than proactive or predictive maintenance strategies. These inefficiencies collectively reduce the return on investment for wind energy projects and hinder the broader adoption of wind power as a reliable energy source.

Therefore, there is a clear and urgent need to develop an efficient, cost-effective, and scalable solution for monitoring wind energy systems in real time. The proposed system addresses these challenges by implementing a cloud-based 10T approach that enables continuous data collection, remote accessibility, real time analysis, and automated alert generation. By replacing outdated and expensive monitoring methods with an intelligent and affordable system, it aims to improve turbine performance, reduce unplanned downtime by up to 30%, lower maintenance costs, and enhance the overall reliability and efficiency of wind energy generation for both large-scale and small-scale installations.

3.2 Objectives of the Project:

Here are 8 project objectives in plain format:

1. To develop a real-time I0T-based data acquisition system that continuously monitors key wind turbine parameters including wind speed, rotor RPM, voltage, current, temperature, and power output.

2. To design a cloud-based centralized data storage platform that securely stores all collected sensor data, enabling long-term performance tracking and historical analysis of wind energy systems.

3. To implement a remote monitoring and visualization dashboard that allows operators to access live and historical turbine data from any location through a web or mobile interface.

4. To create an automated fault detection and alert generation mechanism that notifies operators in real time when sensor readings exceed safe operational thresholds, reducing unplanned downtime by up to 30%.

5. To integrate machine learning algorithms for predictive maintenance by analysing historical sensor data to identify patterns and forecast potential faults before they lead to system failures.

6. To build a cost-effective and scalable monitoring architecture as an affordable alternative to traditional SCADA systems, making efficient wind turbine monitoring accessible for small-scale and rural installations in developing regions.

7. To minimise energy losses and improve overall turbine efficiency by enabling continuous data-driven performance optimisation, targeting the 5-10% annual energy loss caused by inadequate monitoring practices.

8. To enhance the reliability and return on investment of wind energy projects by replacing reactive maintenance approaches with proactive strategies, thereby lowering maintenance costs and supporting broader wind energy adoption globally.

CHAPTER-IV

PROPOSED METHEDOLOGY

Wind Energy Monitoring System – Block Diagram(components):

The diagram represents the working of the wind Energy Monitoring System using a Cloud-Based Approach (**Fig-1: Components Overview**)

1. **Wind Turbine Model**
   The miniature wind turbine is used to simulate wind energy generation in the prototype system. Into electrical energy for monitoring and analysis.

2. **Microcontroller Unit (ESP8266)**
   The ESP8266 microcontroller acts as the main processing unit of the system. It collects sensor data, processes it, and sends the information to the cloud platform for remote monitoring

3. **Power Supply / Battery Unit**
   The power supply unit provides the required electrical power to operate the sensors, controller, and communication modules in the monitoring system.

4. Voltage Sensor Module

The voltage sensor measures the voltage generated by the wind turbine. It helps in monitoring the electrical output and performance of the system.

5. Current Sensor Module

The current sensor monitors the amount of current flowing through the circuit during wind energy generation. It is useful for analyzing power consumption and efficiency.

6. Communication Module (Wi-Fi/GSM)

The communication module enables wireless transmission of real-time data from the monitoring system to cloud servers using Wi-Fi or GSM technology.

7. Connecting Wires and Circuit Board

The wires and circuit board establish electrical connections between all components, ensuring proper communication and power distribution throughout the system.

CHAPTER-V

SOFTWARE REQUIREMENTS :

1. Arduino IDE
2. Embedded / Arduino Language
3. Firebase
4. HTML, CSS, JavaScript
5. Windows OS
6. Serial Monitor

HARDWARE REQUIREMENTS :

1. Wind Turbine Model
2. ESP8266
3. Voltage
4. Current Sensor
5. Power Supply/B
6. Wi-Fi/GSM Module
7. Connecting Wires
8. Breadboard
9. LED Indicators
10. Switches

CHAPTER-VI

SYSTEM IMPLEMENTATION

Model Architecture:

I. Overview

The system follows a layered 10T architecture that connects a physical wind turbine model to a cloud-based analytics and monitoring platform. Data flows continuously from hardware sensors through wireless communication, processing, prediction, storage, and finally to a user-facing dashboard.

II. Architecture Layers

1. Data Acquisition Layer

   Collects real-time data using sensors:

   Wind Speed Sensor, Temperature Sensor, vibration Sensor , Voltage & Current

   Installed on wind turbines to capture environmental and operational data

2. Data Transmission Layer

   Transfers collected data to the processing unit Technologies used:

   Wi-Fi GSM 10T modules (e.g., ESP8266)

   Enables remote monitoring

3. Data Processing Layer

   Preprocesses the data:

   Noise removal Data filtering Normalization

   Microcontroller or cloud performs initial processing

4. Analysis & Prediction Layer

   Applies algorithm to:

   Analyze wind patterns, Predict power output, Detect faults

   Can use:

   Threshold-based analysis

5. Storage Layer

   Stores data in:

   Cloud databases, Local servers, Maintain Historical records for analysis

6. Application Layer (User Interface)

   Displays data via, Web dashboard, Mobile app

   Shows:

   Live wind speed Power generation, Alerts and reports

WIND ENERGY MONITORING SYSTEM ARCHITECTURE

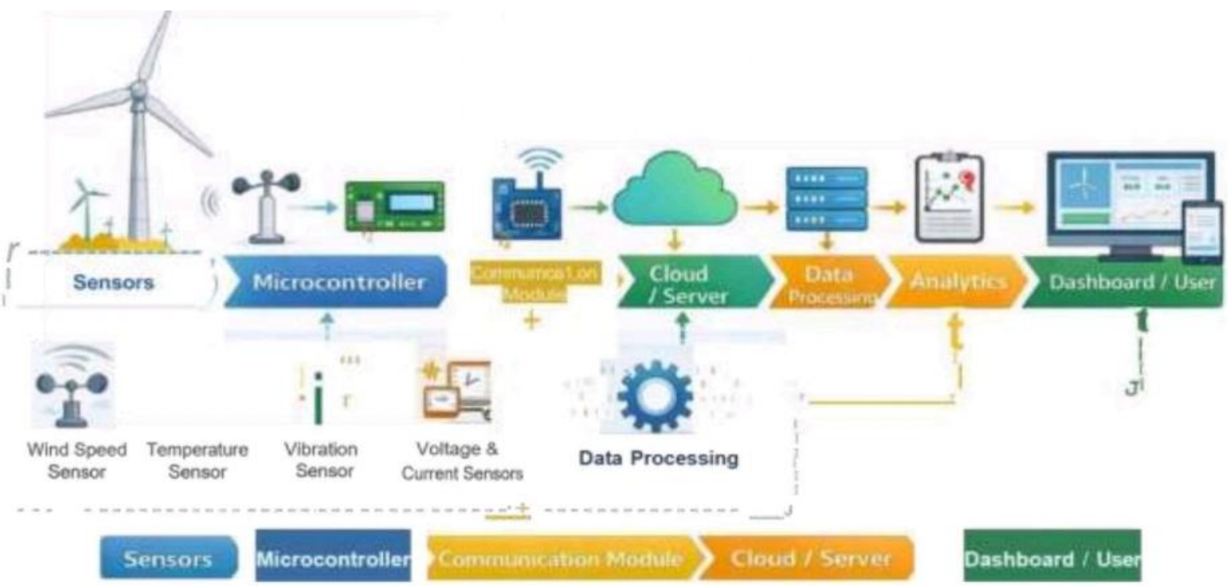

Fig-2: Systematic Architecture

CHAPTER-VII

RESULTS AND DISCUSSION

HARDWARE DEMONSTRATION

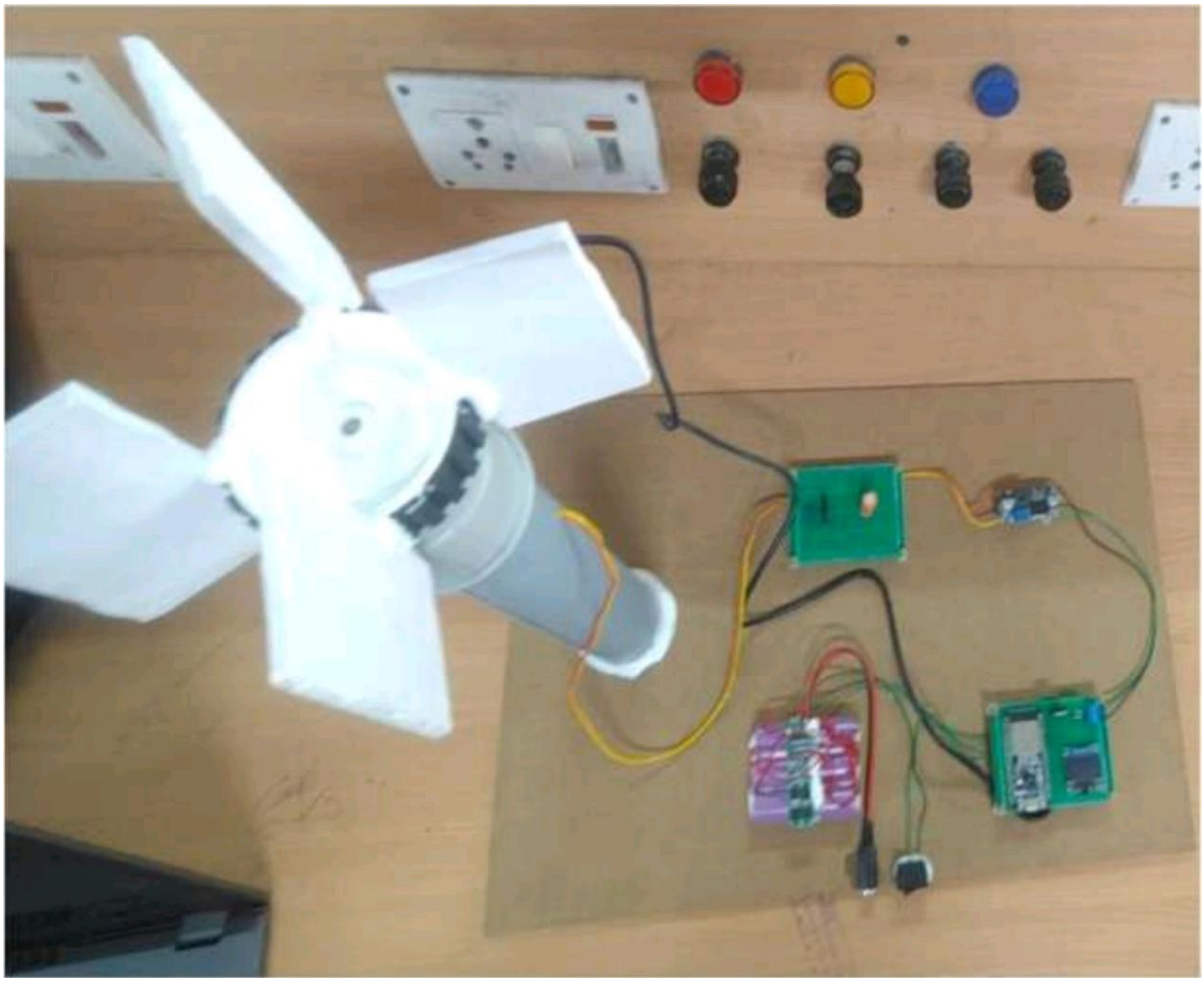

Fig-3: Prototype

The prototype of the Wind Energy Monitoring System is designed to demonstrate real-time monitoring and analysis of wind energy parameters using an IoT-based approach. It consists of a compact setup that integrates sensors, a microcontroller, and communication module to simulate the functioning of a real wind turbine monitoring system.

CHAPTER-VIII

COMMUNITY IMPACT

8.1 Encouraging Equality and Fairness

The Wind Energy Monitoring System promotes equity and fairness by improving access to reliable and affordable energy for all communities, including rural and underserved areas. By enabling efficient monitoring and reducing energy losses, it helps ensure that electricity is distributed more consistently and fairly. The system supports decentralized energy generation, allowing smaller communities to benefit from local wind resources instead of depending entirely on centralized power systems.

Additionally, the use of smart monitoring reduces operational costs, which can contribute to lowering electricity prices over time. This makes renewable energy more accessible to economically weaker sections of society. It also encourages inclusive technological growth by creating opportunities in local employment, skill development, and participation in sustainable energy initiatives.

8.2 Effects on Environment

The Wind Energy Monitoring System has a positive impact on the environment by enhancing the efficiency of renewable energy generation. By optimizing turbine performance and reducing downtime, it increases the use of clean wind energy and reduces reliance on fossil fuels. This leads to a significant decrease in greenhouse gas emissions and helps in mitigating climate change.

Furthermore, efficient monitoring minimizes energy wastage and ensures sustainable utilization of natural resources. The system also supports long-term environmental conservation by promoting clean energy adoption and reducing air pollution. Overall, it contributes to a greener ecosystem and supports global sustainability goals.

CHAPTER-IX

CONCLUSION AND FUTURE SCOPE

CONCLUSION:

The project "Wind Energy Monitoring System: A Cloud-Based Approach" successfully demonstrates the use of modem technologies such as 10T and cloud computing for efficient monitoring of wind energy systems. The proposed system enables real-time data collection, transmission, and analysis of key parameters like wind speed, turbine performance, and power output. By integrating sensors with a microcontroller and cloud platform, the system provides remote accessibility and continuous monitoring, eliminating the need for manual supervision.

The implementation of data visualization tools helps users easily understand system performance, while the alert mechanism ensures quick response to any abnormal conditions. This combination of intelligent monitoring and automated alerting significantly improves operational efficiency, reduces maintenance costs, and enhances the overall reliability of wind energy generation.

Overall, this project proves to be a cost-effective, scalable, and user-friendly solution for modem wind energy management. It also supports sustainable energy utilization by optimizing the performance of renewable energy resources, making it a valuable contribution towards the broader adoption of clean and reliable energy systems.

Overall, this project proves to be a cost-effective, scalable, and user-friendly solution for modem wind energy management. It also supports sustainable energy utilization by optimizing the performance of renewable energy resources, making it a valuable contribution towards the broader adoption of clean and reliable energy systems.

FUTURE SCOPE:

Although the proposed system provides an effective solution for monitoring wind energy systems, there are several areas where further improvements can be made:

l. Integration with Artificial Intelligence (Al):

Advanced Al and machine learning algorithms can be implemented for more accurate predictive maintenance and fault detection.

2. Smart Grid Integration:

The system can be connected to smart grids for better energy distribution, load balancing, and efficient power management.

3. Mobile Application Development:

A dedicated mobile app can be developed with advanced features like real-time alerts, control options, and user-friendly dashboards.

4. Enhanced Data Security:

Implementation of encryption and cybersecurity measures to protect data stored on the cloud.

5. Hybrid Renewable Energy Systems:

The system can be extended to monitor multiple energy sources such as solar and wind together for improved energy efficiency.

7. Automation and Control Features:

Future versions can include automatic control of turbines based on environmental conditions.

8. Use of Advanced Communication Technologies:

Technologies like LoRa and 5G can be used for better connectivity, especially in remote areas.

9. Scalability for Large Wind Farms:

The system can be expanded to monitor multiple turbines across large wind farms with centralized control.

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INTERNAL QUALITY ASSURANCE CELL

PROJECT AUDIT REPORT

This is to certify that the project work entitled "Wind Energy Monitoring System: A Cloud-Based Approach" categorized as an internal project done by **P. HEMALATHA, S. PRANEETHA, G. PRAVALLIKA, R. VAMSI KRISHNA** and **M. BHARGAV RAKESH** of the Department of Computer Science and Engineering, under the guidance of **DR. A. RAMKUMAR** during the Even semester of the academic year 2025 - 2026 are as per the quality guidelines specified by IQAC.

Quality Grade

Deputy Dean (IQAC)

Administrative Quality Assurance **Dean (IQAC)**